

ILD — IPF Treatment & the Antifibrotic Paradigm Shift



1. Grandfather clock

WHAT YOU SEE

A tall grandfather clock standing beside the line of old men — the clock IS the '2014 paradigm shift' theme: antifibrotics slow the clock of FVC decline but cannot turn it back.

WHAT IT ENCODES

IPF is a disease of aging — the median age at diagnosis is ~65, and the 'clock' theme (the 2014 paradigm shift) means antifibrotics buy TIME rather than reverse disease. They decelerate the time-course of FVC decline but cannot undo established honeycomb fibrosis. Think of the whole scene as 'slowing the clock,' not turning it back.

BOARD TRAP

WRONG answer: 'start antifibrotic to restore lung function / wean off oxygen.' Antifibrotics only flatten the decline curve; the discriminator is that no therapy short of transplant improves established fibrosis, so any answer implying reversal or normalization of FVC/DLCO is incorrect.

2. Crowd of older men

WHAT YOU SEE

A long queue of grey-haired elderly men (one marked '60+') — the all-male, aging crowd IS the classic IPF demographic: man over 60, ever-smoker.

WHAT IT ENCODES

Classic IPF demographic: man >60, ever-smoker, presenting with chronic exertional dyspnea, dry cough, and bibasilar 'Velcro' inspiratory crackles, often with digital clubbing. These features plus a UIP pattern on HRCT make IPF the diagnosis without biopsy. The combination of age + smoking + insidious dyspnea is the high-yield trigger.

BOARD TRAP

WRONG answer: labeling a young woman with ILD as 'IPF.' Discriminator — a patient <50, female, or with arthralgias/Raynaud/rash should redirect to CTD-ILD (scleroderma, RA, dermatomyositis with anti-MDA5/Jo-1) or NSIP, which are steroid/immunosuppressant-responsive, not antifibrotic-first.

3. FVC decline slows/reverses (false)

WHAT YOU SEE

Chart 'FVC declining steadily' with note 'slows / reverses'

WHAT IT ENCODES

Both pirfenidone and nintedanib reduce the annual rate of FVC decline by roughly 50% (from ~200 mL/yr toward ~100 mL/yr) in the ASCEND/CAPACITY and INPULSIS trials. The curve still slopes downward — it just slopes less steeply. The 'reverses' label is deliberately the false hook the scene is teaching you to reject.

BOARD TRAP

WRONG answer: 'antifibrotics reverse fibrosis / produce a sustained rise in FVC.' The discriminator is the trial endpoint: the benefit was a SLOWER rate of decline, not improvement; survival benefit is modest and pooled, never a cure.

4. Survival meter

WHAT YOU SEE

Color survival gauge labeled 'survival meter'

WHAT IT ENCODES

Untreated IPF has a grim prognosis — median survival ~3–5 years from diagnosis, worse than many cancers. Poor-prognosis markers: lower baseline FVC/DLCO, a >10% absolute FVC drop over 6–12 months, pulmonary hypertension, and acute exacerbations. Antifibrotics give only a modest mortality/progression benefit.

BOARD TRAP

WRONG answer: quoting a benign prognosis or 'watchful waiting indefinitely.' The discriminator is that confirmed UIP/IPF warrants early antifibrotic initiation and transplant referral, because the survival meter runs low fast — delay costs lung function that never returns.

5. Pirfenidone (PIRFEN) figure

WHAT YOU SEE

Woman labeled PIRFEN holding a wrench, 'slows progression'

WHAT IT ENCODES

Pirfenidone is an oral antifibrotic (antagonizes TGF-β and reduces fibroblast proliferation/collagen) dosed by titration up to 801 mg three times daily with food. It slows FVC decline in mild-to-moderate IPF (ASCEND/CAPACITY) and requires LFT monitoring. It is an IPF/progressive-fibrosis drug, not an anti-inflammatory.

BOARD TRAP

WRONG answer: 'give pirfenidone for steroid-responsive COP or cellular NSIP.' Discriminator — those inflammatory ILDs respond to corticosteroids; pirfenidone is reserved for the fibrotic UIP phenotype, so the radiographic/histologic pattern decides the drug.

6. Pirfenidone photosensitivity

WHAT YOU SEE

A bright sun/sunburst glaring on the PIRFEN figure — the sun IS pirfenidone's signature photosensitivity rash (counsel sun protection).

WHAT IT ENCODES

Pirfenidone's signature adverse effects: PHOTOSENSITIVITY rash (counsel strict sun protection/sunscreen), GI upset (nausea, anorexia, dyspepsia), and elevated transaminases. The sun icon is the memory hook distinguishing it from nintedanib's GI/bleeding profile. Avoid strong CYP1A2 inhibitors (e.g., fluvoxamine) and smoking, which alter levels.

BOARD TRAP

WRONG answer: attributing a sun-exposed rash on pirfenidone to a new CTD/dermatomyositis flare. Discriminator — drug-induced photosensitivity resolves with sun avoidance and dose adjustment, whereas a true CTD rash comes with myositis, mechanic's hands, or autoantibodies.

7. Nintedanib (NINTE) figure

WHAT YOU SEE

Man labeled NINTE with wrench, 'slows FVC decline'

WHAT IT ENCODES

Nintedanib is an oral intracellular tyrosine kinase inhibitor (blocks VEGFR, FGFR, PDGFR) dosed 150 mg twice daily; it slows FVC decline in IPF (INPULSIS), in scleroderma-ILD (SENSCIS), and in progressive fibrosing ILD (INBUILD). Its anti-angiogenic action explains both its antifibrotic effect and its bleeding risk.

BOARD TRAP

WRONG answer: 'use nintedanib as the immunosuppressant for active inflammatory/cellular ILD.' Discriminator — nintedanib is antifibrotic, not immunosuppressive; purely inflammatory ILD needs steroids ± mycophenolate, and the UIP/progressive-fibrosis phenotype is what selects nintedanib.

8. Nintedanib diarrhea

WHAT YOU SEE

A burst of fire/flames erupting beside the NINTE figure — the burning gut IS nintedanib's hallmark diarrhea (>60% of patients).

WHAT IT ENCODES

Nintedanib's hallmark adverse effect is DIARRHEA (>60% of patients — manage with loperamide and hydration), plus nausea and hepatotoxicity requiring baseline and periodic LFT monitoring. Contrast with pirfenidone's photosensitivity. The 'fire/heat' near NINTE is the diarrhea cue.

BOARD TRAP

WRONG answer: discontinuing nintedanib at the first loose stool. Discriminator — diarrhea is usually manageable with antimotility agents and dose reduction; true stopping triggers are ALT/AST >3–5× ULN (hepatotoxicity) or significant bleeding, not mild GI symptoms.

9. Increased bleeding/PE risk

WHAT YOU SEE

Red clot icons under both drug figures labelled 'increased PE risk' — the clots ARE nintedanib's anti-VEGF bleeding plus arterial thromboembolic/MI risk.

WHAT IT ENCODES

Nintedanib's anti-VEGF activity increases BLEEDING risk and, via impaired vascular healing, raises arterial thromboembolic/MI risk and risk of GI perforation. Use caution in patients on full-dose anticoagulation and in those with recent cardiovascular events. This is the key safety nuance unique to nintedanib over pirfenidone.

BOARD TRAP

WRONG answer: starting full anticoagulation plus nintedanib in an IPF patient without weighing risk. Discriminator — the combination amplifies bleeding; the question usually wants you to recognize the heightened bleeding/cardiovascular risk rather than reflexively anticoagulate.

10. PPI / GERD acid

WHAT YOU SEE

Tipped PPI bottle, 'GERD/reflux leaking acid'

WHAT IT ENCODES

GERD is highly prevalent in IPF, and chronic micro-aspiration is thought to contribute to repetitive alveolar injury driving fibrosis. Guidelines suggest considering anti-reflux therapy (PPI) as adjunctive management, though evidence for slowing fibrosis is weak. The reflux–fibrosis link is the testable association.

BOARD TRAP

WRONG answer: 'PPI therapy is a proven disease-modifying treatment that halts IPF progression.' Discriminator — PPIs are adjunctive/symptomatic for GERD; only antifibrotics slow decline and only transplant is definitive, so PPI is never the primary IPF therapy.

11. Fire / acute exacerbation

WHAT YOU SEE

A huge fireball exploding behind the drug figures — the sudden conflagration IS an acute exacerbation of IPF (abrupt new bilateral GGO, >50% mortality).

WHAT IT ENCODES

Acute exacerbation of IPF = acute (<1 month) unexplained worsening of dyspnea with NEW bilateral ground-glass/consolidation superimposed on the UIP background, after excluding infection, PE, heart failure, and fluid overload. It carries very high in-hospital mortality (often >50%); management is supportive ± high-dose steroids (low-quality evidence).

BOARD TRAP

WRONG answer: 'treat the acute decompensation as simple CHF/diurese and discharge.' Discriminator — you must rule out HF, PE, and infection, but new diffuse GGO on a UIP lung with no other cause is an acute exacerbation, an ominous event, not volume overload.

12. Advanced IPF lung module

WHAT YOU SEE

Hanging lung sign 'advanced IPF lung module', 'last resort'

WHAT IT ENCODES

Lung TRANSPLANTATION is the only definitive, survival-prolonging therapy for end-stage IPF and should be referred early (at diagnosis of advanced/progressive disease, before patients are too deconditioned). Antifibrotics buy time to transplant; supplemental O2 and pulmonary rehab are supportive adjuncts.

BOARD TRAP

WRONG answer: 'high-dose corticosteroids' or 'cyclophosphamide' for end-stage IPF. Discriminator — immunosuppression neither helps nor is definitive (the PANTHER trial showed prednisone/azathioprine/NAC INCREASED death/hospitalization in IPF); transplant is the only definitive option.

13. CTD-ILD only meds (X)

WHAT YOU SEE

Shelf 'CTD-ILD only' with bottles marked red X

WHAT IT ENCODES

Immunosuppressants (mycophenolate mofetil, azathioprine, cyclophosphamide, rituximab) are for CTD-associated ILD and inflammatory NSIP — NOT idiopathic UIP/IPF. The red X means 'do not use these in IPF.' Pattern recognition: inflammatory/cellular = immunosuppress; fibrotic/UIP = antifibrose.

BOARD TRAP

WRONG answer: 'mycophenolate or azathioprine for idiopathic UIP/IPF.' Discriminator — the landmark PANTHER-IPF trial showed combination immunosuppression (prednisone + azathioprine + NAC) caused HARM; immunosuppression is correct only when there is an underlying CTD or cellular inflammatory pattern.

14. Mortality + morbidity tombstone

WHAT YOU SEE

A grey gravestone reading 'mortality + morbidity' — the tombstone IS the grim ~3–5 yr median survival that persists even on antifibrotics.

WHAT IT ENCODES

Even on antifibrotics, IPF remains relentlessly progressive with high morbidity and mortality; median survival is only ~3–5 years and the main drivers of death are progressive respiratory failure and acute exacerbations. The tombstone underscores that disease modification is partial and transplant is the only path to longer survival.

BOARD TRAP

WRONG answer: telling a patient antifibrotics will 'cure' or 'normalize life expectancy.' Discriminator — counseling must reflect a progressive, fatal disease with modest pharmacologic benefit; over-optimistic prognosis is the trap and palliative/transplant planning is appropriate.